

Exploring Early Adopters' Use of AI Driven Multi-Agent Systems to Inform Human-Agent Interaction Design: Insights from Industry Practice

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Abstract

This case study explores the experiences of Microsoft employees, who are early adopters of multi-agent generative AI systems, as they experiment with these technologies to design, test, and deploy new tools attempting to bridge the gap between existing Microsoft products and emerging AI capabilities. Thirteen developers and creators participated in 60-minute semi-structured interviews to elicit their challenges, use cases, and lessons learned from their experimentation with multi-agent AI frameworks. A thematic qualitative analysis process was conducted to analyze the interview data. Participants reported building multi-agent AI tools to address tasks in team collaboration, productivity, customer support, creative processes, and security. Strategies for managing complexity, enhancing transparency, and balancing agent autonomy with human oversight were found to be important human-agent interaction design considerations. Findings from this study highlight the capabilities and limitations of specialized multi-agents within the contexts of participants' use cases and provide insights to inform the human-agent interaction design of future multi-agent generative AI systems.

CCS Concepts

• **Human-centered computing** → **Empirical studies in HCI**; **Empirical studies in interaction design**; *Natural language interfaces*.

Keywords

Artificial Intelligence, User Experience, Challenges and Limitations of AI Use

*This work was done during internship at Microsoft Research

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1 Introduction

The rise of multi-agent AI systems, leveraging Large Language Models (LLMs), called “Multi-agent Generative AI system” offers promising solutions to navigate the complex digital user experience ecosystem of applications and platforms to accomplish personal and work-related tasks via digital devices. A multi-agent AI tool refers to LLM applications composed of multiple autonomous conversable agents that interact and cooperate to achieve individual or collective goals. These tools are typically created using frameworks such as AutoGen, LlamaAgents, and LangGraph.

Multi-agent generative AI systems, wherein multiple agents collaborate to conduct tasks on behalf of a human, represents a technological shift in the design and development of intelligent systems from single (co-pilot or assistant AI) chatbot interfaces to more complex networks of agents interacting with other agents and humans. However, being able to create user experiences to communicate agentic AI actions and their interactions with humans requires understanding the computational design of agents, the infrastructure that connects them, and the human operators' mental models about how the technology functions.

Human computer interaction (HCI) research emphasizes human-centered design to ensure usability, trust, and alignment with user needs, yet challenges remain, particularly in the context of multi-agent AI systems where user control and transparency are often limited [11]. Traditional participatory design models have been suggested for AI [1, 13], and there is an opportunity now to apply such approaches to multi-agent AI systems. Experiences of early adopters, who are also technology practitioners, can help bridge the gap between system capabilities and user expectations, preventing issues like misaligned mental models, which can lead

to over-reliance or skepticism, undermining effective human-AI collaboration [5, 6, 8, 9, 14].

Early adopters are experimenting with multi-agent generative AI to create novel HCI system models to perform digital tasks on behalf of the user. For example, the Rabbit R1 capabilities enable users to ask (via speech) an AI agent to perform a task on their behalf. The Rabbit R1 leverages apps and services, such as Uber and DoorDash, to perform the digital steps necessary to order food delivery or request a ride to the airport automatically, without requiring the user to open an app. These early adopters can provide user feedback on how to design and address the complexities of multi-agent AI systems, where issues like agent autonomy, transparency in decision-making, and user control present significant challenges. How early adopters interact, and address challenges of new technological innovation can highlight potential struggles non-experts may face [3], providing AI designers with insights on how to balance agent autonomy with human control.

Therefore, we can learn from early adopters' interactions with multi-agent generative AI frameworks to better understand HCI requirements, such as usability and transparency. We can assess multi-agents' technological capabilities and limitations by examining the experiences of these early adopters and their experimentation with multi-agent generative AI applications. Our case study aims to explore what early adopters (i.e., developers, creators) are building with multi-agent generative AI systems to inform the human-agent interaction design of future multi-agent AI tools.

2 Methodology

We conducted 60-minute semi-structured interviews with 13 early adopters (creators and developers) of multi-agent AI frameworks from Microsoft. This study took place in the Summer of 2024, in the middle of rapid adoption of AI tools in all sectors. During this moment in time, AI developers across many technology companies and organizations were learning quickly about capabilities and limitations as they attempted to innovate or integrate AI in their respective products and services. We asked participants questions related to their experiences building multi-agent generative AI solutions. We conducted a qualitative analysis of the transcripts from the interviews to understand what capabilities and limitations/challenges they encountered while building multi-agent AI tools within the context of their use cases and jobs to be done of agents. The following sections provide additional details on our participant recruitment process, informed consent, analysis, and participant demographics.

Recruitment and Informed Consent Process: We recruited study participants using a combination of Microsoft-internal community distribution email lists and snowball sampling techniques. The primary inclusion criteria for participation required individuals have first-hand experience creating or experimenting with a multi-agent AI framework within their professional roles and possess an intermediate or greater level of knowledge about multi-agent generative AI. Inclusion criteria for this study required participants to be 18 years of age or older. Participants were excluded if they had higher executive roles in industry, to avoid any potential bias from senior decision-makers.

Participants interested in the study were sent a Microsoft Forms survey to assess their eligibility based on the study criteria and to review the study informed consent. If they met the eligibility criteria requirements and agreed to the terms outlined in the informed consent, they were sent a Microsoft Teams meeting link to schedule a 60-minute interview. This study was reviewed and approved by Microsoft Research Institutional Review Board (ID10935).

Participants and Interview Process: The study involved 13 participants who volunteered to participate in our study. These participants represented a diverse range of functional roles within Microsoft. Of the 13 participants, six held non-technical job roles, while the remaining seven served technical job functions. Participants were primarily located in the United States, Europe, and the UK.

The semi-structured interviews were designed to explore participants' reflections on tasks involving multi-agent AI frameworks, as well as their understanding of such systems. The interview questions aimed to capture participants' explanations and reasoning for using multi-agent AI systems¹, their approaches to transparency in system development², and any challenges they encountered³. The interview questions were further framed within the concept of Explicit Explanation, as outlined by Hoffman et al. [7], which examines how users understand, interpret, and explain the systems they develop or interact with⁴. The open-ended nature of the questions allowed an in-depth exploration of each participant's unique use case⁵, offering rich insights into the early adoption of multi-agent generative AI systems.

All participants self-reported that they had either developed a proof of concept (POC) or an actual multi-agent generative AI tool for personal or work-related use, or were in the process of creating one. These tools were built using various frameworks, including but not limited to AutoGen [4, 15], Semantic Kernel, TaskWeaver [10], BizChat, Sydney, LlamaIndex, Oagent, and several custom-built frameworks. They reported that the primary motivations for developing these tools were to simplify their work, learn about the technology, understand its capabilities and limitations, experiment, and share knowledge with others.

Analysis: We employed collaborative affinity mapping to identify and categorize common themes, patterns, and relationships between concepts and keywords that stood out to us in the data. We applied a thematic analysis approach [2] to the interview transcripts, which allowed us to construct key insights into early adopters' development of multi-agent AI tools, including their perceived capabilities, limitations, and design preferences. The results of this analysis are discussed in detail in the following section.

3 Findings

In this section we present the themes we found across our interviews with early adopters of multi-agent generative AI systems. We start by outlining the different use-cases, tasks, and specialized

¹e.g., "When and why do you use multi-agent AI tools instead of single-agent AI?"

²e.g., "What comes to your mind when I say transparency practices in multi-agent AI?"

³e.g., "Based on your experience, can you tell what are some capabilities and limitations of using this tool?"

⁴e.g., "Based on your knowledge, can you describe multi-agent AI?"

⁵e.g., "Without showing me any business confidential projects, can you walk me through the tool showing how you use it in a day-to-day task?"

agents that study participants developed. Next, we list the common patterns found in how participants described the capabilities and limitations of these multi-agent generative AI tools. We end the section with an overview of the design preferences and lessons learned from these early adopter participants.

3.1 Building with Multi-agent Generative AI tools: Different Use-Cases

Participants reported using multi-agent generative AI tools across multiple domains to streamline workflows, automate processes, engage in creative tasks, and enhance collaboration. These tools were typically designed to distribute tasks across multiple specialized agents. The use cases for these systems spanned team collaboration, productivity support, creative processes, customer service, and security analysis. Table 1 summarizes the various use cases described by participants, along with the specialized agents they developed for each task.

3.1.1 Use case 1: Team Collaboration and Simulation. An example of multi-agent AI tools in action is their use in simulating team collaboration within a travel agency context. Participants P09 and P07 designed agents that coordinated tasks such as booking flights, planning tours, and answering customer queries. The following specialized agents were created: *Travel Agent (User Proxy)*: the main coordinating agent that gathered user preferences and distributed tasks to other agents. *Ticket Manager agent*: Responsible for finding and booking flights that matched user specifications. *Event agent*: Identified local events during the user's stay and made necessary reservations.

This setup allowed users to simulate real-world travel agency scenarios, optimizing efficiency and improving customer service through seamless interaction between agents by handling complex tasks. P09 noted that this configuration “*mimicked human interactions well and freed up our time from manual work*”. The purpose of building this tool was to learn to use this new technology and gauge the efficiency of the tool.

3.1.2 Use case 2: Productivity and Coding Support. Another significant application of multi-agent systems was observed in productivity and coding support, where participants P02 and P03 used the technology to automate tasks such as generating release notes and newsletters. In the case of **release notes automation**, P02 automated the process of retrieving data from GitHub, summarizing issues, and generating the release notes. The system relied on agents like *user proxy agent*, which allowed the user to create queries and instruct the system to retrieve and summarize relevant issues. The *issue-helper agent* synthesized GitHub issues to streamline the release note generation process, which P02 noted “*the process is so time consuming, so this tool is something that you can use to help yourself*”.

Similarly, in **newsletter automation**, P03 used multi-agent systems to streamline gathering updates from team members, compiling them into a newsletter, and distributing the final output. Agents such as the *teammate outreach agent* collected content from team members, while the *content aggregation agent* compiled the input into a cohesive newsletter. Finally, the *newsletter sending agent* ensured the timely distribution of the newsletter to recipients.

P03 emphasized the system's efficiency, remarking “*I didn't have to chase [anyone from the team] anymore for updates, the agents took care of it all*”.

3.1.3 Use case 3: Customer Support and Service. In the domain of customer support and service, participants P05 and P11 integrated multi-agent systems into existing platforms like Teams and GitHub to streamline issue resolution and project management. For example, in **on-call engineer automation**, P05 used a multi-agent system integrated within Teams to provide daily summaries of pipeline and business intelligence (BI) reports. This system assisted engineers with troubleshooting by generating specific solutions, utilizing the *user proxy agent* to interact with users and deliver final results, and the *on-call agent*, which generated and interpreted code to aid in issue resolution.

P11 implemented multi-agent systems in **GitHub flow for development projects** to manage development cycle tasks such as project planning and submitting pull requests. The *PM agent* helped create a project plan based on user input, while the *development lead agent* decomposed tasks into smaller components and assigned them to developers. Once the tasks were completed, the *pull request manager agent* finalized and submitted the pull request. P11 said this system “*handled everything from planning to execution, reducing human intervention and speeding up the entire process*”.

3.1.4 Use case 4: Creative Workflow and Learning. Generative AI has proven to be an effective tool for creative endeavors including content generation, and refinement. From the beginning of its adoption, generative AI has been valued for enabling brainstorming particularly for blank page scenarios. Four of the participants shared how they've used agent systems to chain these generative AI capabilities together to enable creative workflows whilst developing human-agent collaboration capabilities to guide users through complex creative projects.

For example, the prospect of writing a book is a daunting task that requires sense making, outlining, narrative structuring, and ongoing revising and refining. P13 explained that an agent system can be a book writing companion, guiding a user through the process step by step. P13's **book generation agent system** includes a *chapter structuring agent* that guides users through content structuring into book chapters. And then a *content refinement agent* does the work of editing the user's content iteratively at each stage of the writing process. And an image generating agent provides users the ability to create images that match the book's prose.

Similar to authoring a book, marketing campaigns are another complex, multi-step project that have discreet specialized tasks that build on one another towards completion. P11 developed a **marketing campaign generator** that includes role-based agents that contribute to the various deliverables that go into a marketing campaign, including, a *content writing agent*, *social media persona agent*, *graphic designer agent*, and an *auditor agent* that ensure all the campaign content complies with brand guidelines and legal requirements. Since this is a system, the agents work together to build off the content each has generated, optimizing for the various mediums and channels.

Participants developing agent systems for creative workflows identified the ability to have both specialized agents and agents that can be reusable to do repetitive tasks like generating, synthesizing,

Table 1: Participants' Use-cases of multi-agent GenAI tools and specialized agents created task wise

Use Case Scenario	P's	Specialized Agents	Tool Description
Team collaboration and Simulation	P09, P07	Travel Agent (User Proxy), Ticket Manager Agent, Event Agent	Travel agency simulation: Used for simulating a travel agency, coordinating tasks such as flight bookings, tour planning, and responding to customer queries.
Productivity and coding support	P03	Teammate outreach agent, Content aggregation agent, Newsletter sending agent	Newsletter Automation: Collected input from team members, compiled newsletters, and distributed them to recipients.
Productivity and coding support	P02	User proxy agent, gpt4o (System agent), Issue-helper agent	Release Notes Automation: Automated the retrieval of GitHub issues based on the user's query and summarizes to create release notes.
Customer support and service	P05	User proxy agent, Writer/On-call agent	On-call engineer automation: Integrated within Teams to provide daily issue summaries and troubleshooting for system operations.
Customer support and service	P11	Project Manager agent, Development Lead agent, Developer agent, Pull request manager agent	GitHub Flow Management: Managed tasks within a GitHub development cycle, from project planning to submitting pull requests.
Creative work and learning	P07	Joke creation agent, Feedback monitoring agent, Learning agent	Joke Generation: Generated and refined jokes based on user preferences and feedback.
Creative work and learning	P14	Video summarizer agent, Content creation agent	YouTube Video Summarization: Summarized video content and generated social media posts and blog content.
Creative work and learning	P13	Chapter structuring agent, Content refinement agent, Image creation agent	Book Generation: Guided the user through structuring and refining book chapters, while also assisting with image creation.
Creative work and learning	P11	Content writer agent, Graphic designer agent, Social media persona agent, Auditor agent	Marketing Campaign Generation: Automated copywriting, graphic design, and legal compliance for marketing campaigns.
Security	P06	Data Scientist agent, QR code reader, Google assistant, Summarizer agent	QR code phishing detection: Analyzed suspicious emails and decoded QR codes to detect phishing attempts.
Security	P06, P14	Regular analyst agent, Senior analyst agent, Writer agent, Editor agent, Threat intelligence analyst agent, After-action investigative analyst agent	Phishing email triage: Analyzed phishing emails and produced comprehensive reports on potential security threats.

and refining content as crucial system capabilities. They expected the content to be generated across different formats (e.g., social media, video, image, text), extensible across platforms, and the agent system to make iterative improvements based on user feedback.

3.1.5 Use case 5: Security. Digital security is an industry sector that requires ongoing detection management where automation greatly improves effectiveness against security breaches. According to P06 and P14, agent systems have the potential to optimize detection management. P06 has developed a **QR code phishing detector** that analyzes emails for suspicious QR codes, decodes them, checks them for malicious links, and generates security reports. For this workflow, they used a mix of role-based and task-based agents, a *data scientist agent* that analyses the input data, a *web search agent* to verify the URL, a *QR code agent* that extracts the link from the QR code and follows the link to see if it leads to a suspicious site, an *image agent* that analyzes the email's images and extracts data

for further investigation, and a *summarizer agent* that combines all the other agents' conclusion into a report summary.

P14 described a similar system for **phishing email triage**; however, the agents were all role-based with specializations to execute specific tasks in a workflow which follows industry security analyst workflow patterns. The analyst agent reviews the email and gathers initial indicators or artifacts (e.g., suspicious email addresses, links, etc.) for further investigation. The *senior analyst agent* reviews the work done by the *analyst agent* and critiques or enhances it. The *writer agent* produces the written report summarizing the findings from the phishing triage. The *editor agent* then critiques the report, identifying areas for improvement, and either approves the report or sends it back for revisions. The *threat intelligence analyst agent* provides deeper analysis of threat-related information, adding intelligence on potential risks or attack vectors associated with the phishing email. And finally, a *post-mortem agent* conducts a post-investigation review to identify lessons learned.

In both cases, one can see how the security analyst flow has been methodically broken down into tasks that are conducted by different agents, and specialized agents are incorporated in to do the discreet tasks done by human disciplines like data science. P06 and P14 identified these systems' capabilities as an automated workflow of what LLMs can be capable of providing: analyzing data sets, decoding, and generating reports. They expect this system to have a rapid accurate detection with minimal effort to the end user.

3.2 Capabilities of the Multi-agent AI tools

The creations of early adopters highlighted several key capabilities of multi-agent AI systems. In each of the use cases outlined in the above section, the tools developed exhibited common capabilities, as detailed in Table 2, which facilitated the alignment between system expectations and user expectations. The capabilities include specialized agents for specific tasks, automation of the complex workflow, orchestration of the agents, processing and synthesizing of large amount of information, as well as integration with multiple external systems using agents connected to APIs.

Task Specialization: Participants created agents with specialized roles to improve efficiency and effectiveness in task execution. P06 mentioned specialized agents in security use-case: *"Agents like the data scientist, Google assistant, critic, QR code reader, and then a final decision-making agent, each of those tasks is assigned to a different agent."*

Complex Task Automation: These multi-agent systems are automating workflows that go beyond simple question-answering, handling intricate, multi-step processes. These systems handle complex sequences of actions, reducing manual effort. P14 described automating social media content creation use-case: *"I have that writer, take this transcript, pull out all the themes, and write a blog post about it, and then also create a tweet thread. Then I pass that into an image creator. At the end, it'll give me the image, the tweet thread, and the blog post."*

Orchestration of Agents: These multi-agent tools demonstrate the ability to decompose complex tasks into manageable subtasks, orchestrating specialized agents to handle each component efficiently. This is done by delegating specific responsibilities to agents best suited for each task. Participant P02 highlighted this advantage in Release notes automation use case: *"What I can foresee the multi-agent can do is that you just create a task and there will be some orchestration strategy for you to collaborate those agents, so you don't need to orchestrate the questions to a specific agent manually."*

Handling Large Amount of Information: The ability of these LLM agents to handle vast data efficiently was highlighted by many when discussing their use cases. Use of different agents to break down the tasks increases the memory of such system to process large data effectively. P14 states for email triaging and phishing use-case: *"I think again the beauty of these agents is that it takes a massive amount of text, and synthesize it down."*

Integration with External Systems Using Agent Connected to APIs: Agents' ability to interact with external systems through connected APIs expands their functionality and real world applicability. P14 highlighted their capability for the YouTube summarizer use-case as *"You can pull information from the web using the agent."*

If anything has an API, you can call it, and then that API is connected to something in the physical world, then it can actually do something."

These capabilities illustrate the potential of multi-agent systems to solve complex problems, but they also raise questions about usability and transparency for non-expert users, highlighting the need for better user-centered design of these AI tools that align with human workflows and expectations.

3.3 Challenges and Limitations Revealed by Early Adopters

While multi-agent AI systems offer advanced capabilities, early adopters are also facing significant challenges that hinder broader adoption. These challenges, as mentioned in Table 2 use-case wise, include issues related to scalability and system complexity, a lack of transparency and trust, the occurrence of unproductive loops, and unpredictability and error propagation.

Scalability and Complexity: As the number of agents increases, developers struggle with managing workflows, tracking decision-making processes, and ensuring predictable outcomes. The complexity of multi-agent interactions leads to difficulties in debugging and error correction. P03 states *"As you start to grow in number of agents, it may be difficult to sort down the workflows."*

Transparency and Trust: The "black box" nature of LLMs makes it hard for users to understand agent actions, identify error propagation, or assess system reliability. P06 mentions *"Inability to get good traceability or explainability into what the agents are doing is a challenge."* Early adopters stress the need for more transparency tools to foster trust and facilitate debugging. P04 remarks *"You're gonna need more visual tooling for developers to know what's going on."* These limitations along with challenges in testing, debugging, and evaluation of these multi-agent tools, makes it hard to ensure reliability. P04 also noted, *"Debugging software is already hard enough and now we are adding things that can decide when they want to go in different directions."*

Unpredictability and Error Propagation: Due to the probabilistic nature of AI models, early adopters report frequent instances of agents producing errors or "hallucinating" incorrect information, which then propagates across agents and creates larger problems. P01 states: *"Even within that they start to forget things, hallucinate too much."* The negative consequences of these hallucinations can scale out of hand when one agent's error feeds into another agent, leading to compounded inaccuracies. P08 states *"You have thing feeding into thing, what are the odds that's not going to hallucinate?"*

Agents Entering Unproductive Loops: Agents in this multi-agent system can sometimes engage in unproductive loops, continuously interacting without making progress or adding value to the task at hand. This inefficiency is highlighted by participant P06 as *"You get into loops where agents will continue to talk, but not add anything [new] material."*

4 Discussion

The findings from this study underscore the complexities in developing multi-agent generative AI systems, emphasizing the critical need for human-centered design approaches. We present key lessons based on the limitations of using multi-agent system and

Table 2: System Capabilities and Limitations Across Use Cases

Use Case	System Capabilities	System Limitations
Team Collaboration and Simulation (P09, P07)	Task Specialization: Agents handle specific tasks (e.g., bookings). Complex Automation: Automates travel bookings, tour planning, and customer responses.	Complexity: Managing multiple agents for tasks like flight bookings and customer queries becomes difficult as the system scales. Error Propagation: Errors in booking or itinerary creation can cascade into other agents, leading to incorrect responses.
Productivity and Coding Support (P02, P03)	Orchestration: Agents manage specific parts of workflows like newsletters and release notes. Complex Automation: Automates newsletter creation and GitHub issue summaries.	Complexity: Handling many agents (e.g., for newsletter and release notes automation) complicates workflow management. Transparency: Limited traceability in summarizing issue data. Unpredictability and Error Propagation: Incorrect summaries can spread across agents.
Customer Support and Service (P05, P11)	Task Specialization: Agents handle troubleshooting and GitHub task management. Handling Large Information: Efficiently processes system data and GitHub issues. API Integration: Agents integrate with Teams and GitHub.	Latency: Delays in retrieving issue summaries and handling GitHub workflows. Error Propagation: Misleading troubleshooting or pull request errors can escalate. External System Through API: Coordinating multiple agents for making API calls to GitHub and Azure can be complicated.
Creative Workflow and Learning (P07, P14, P13)	Task Specialization: Agents manage creative tasks (e.g., joke generation, video summarization). Complex Automation: Automates multi-step creative processes like marketing campaigns and content creation. API Integration: Accesses web data for content tasks.	Unproductive Loops: Repetitive content generation (e.g., jokes). Error Propagation: Inaccurate video summaries or book chapter creation can affect outputs. Costs: High resource demands for tasks like video summarization.
Security Tasks (P06, P14)	Task Specialization: Agents specialize in security tasks (e.g., phishing detection). Handling Large Information: Processes large volumes of security data. API Integration: Connects to external systems for phishing detection.	Transparency: Hard to assess if phishing detection is accurate. Costs: High computational resources required for analysis with so many agents. Complexity: Tracking multiple agents analyzing threats is difficult.

suggested future directions for fostering user-friendly and reliable multi-agent generative AI tools.

4.1 Key Lessons for Human-centered Design from Early Adopters

The complexity and unpredictability of multi-agent interactions, combined with issues of transparency, error propagation, and unproductive loops, highlight the need for a thoughtful design approach.

Managing Scalability and Complexity: Early adopters noted that increasing the number of agents complicates workflow management. To address this, systems should include *orchestration mechanisms* that automatically delegate tasks, reducing user burden, *Simplified interfaces* for reduced cognitive load along with adaptable transparency options will help users balance ease of use with the ability to access detailed system information when needed. Identifying risk mitigation for error propagation, including taxonomy for risk levels is crucial.

Enhancing Transparency and Trust: Transparency is critical for building trust in agent operations. Many early adopters struggle with understanding “black box” decisions. *Adaptive transparency*

mechanism [12], offering deep insights for technical users and high-level overviews for non-technical users can help build trust by improving error detection and system accuracy.

Balancing Autonomy with Human Oversight: While agent autonomy reduces manual workload, early adopters stressed the need for human oversight in critical decision points, especially in unpredictable scenarios. Integrating *human-in-the-loop mechanisms* [13] where users can intervene in key decisions will ensure greater reliability and user control. Thus, multi-agent systems must incorporate fail-safes that allow human intervention when agents produce uncertain or erroneous outputs.

Preventing Unproductive Loops and Ensuring Efficiency: Unproductive agent interactions, where agents endlessly engage without progress is another limitation of these tools. Due to the interconnected nature of these agentic systems, tracking error propagation resulting from any form of agent failure presents significant challenges. Participant P01 proposed the implementation of a “Watchman agent” as a potential solution. This agent would monitor the activities and interactions between all agents and provide users with an accessible and comprehensible overview of system operations. Such an approach offers a practical safeguard by presenting synthesized insights instead of overwhelming users with raw logs

of the machine learning processes. Early adopters should incorporate such design mechanisms as safeguards to detect and terminate such loops, improving productivity and agent performance.

4.2 Future Directions: Co-Design and Iterative Development

Building on the lessons learned from early adopters and taking inspiration from participatory design model [1, 13, 14], we argue that there is a strong need for future research that focuses on the development of co-designed, context-aware toolkit/frameworks for multi-agent systems, informing the future direction of AI-UX design. Involving both technical and non-technical users (including users with less AI literacy) in iterative development cycles will optimize these systems for real-world needs and allow for continuous refinement based on feedback. Incorporating human-centered design principles, such as adaptive transparency, simplified interfaces, and user control will enhance the alignment of system capabilities with user needs, thereby improving efficiency and fostering trust. Given the current lack of established standards, a human-centered approach is essential to ensure seamless integration of multi-agent generative AI tools into existing collaborative workflows. Without this alignment, the risk of unintended consequences, including reduced user trust and ineffective system use, remains high. As multi-agent systems are more complex than single-agent models, a human-centered design is crucial to mitigating risk.

5 Conclusion

Our case study of early adopters using and developing multi-agent AI systems offers a rich source of insights for human-centered design of complex AI technologies. By exploring the varied use-cases of multi-agent generative AI tools, their capabilities and the limitations/challenges that early adopters have faced, our study highlights the critical need for human-agent design practices that prioritize usability, transparency, and adaptability. The increasing complexity of workflows necessitates orchestration mechanisms to streamline task delegation, complemented by simplified interfaces and adaptable transparency to manage cognitive load. Transparency, essential for trust, can be enhanced through adaptive mechanisms that offer technical insights for experts and high-level overviews for general users, addressing concerns about error detection and propagation. To mitigate unproductive agent loops or error propagation due to agent failures, design mechanisms or safeguards like a “Watchman agent” could be incorporated to monitor interactions, improving system efficiency and usability. Also, balancing agent autonomy with human oversight through human-in-the-loop mechanisms ensures reliability in unpredictable scenarios, empowering users to intervene at critical decision points.

While these findings offer practical design recommendations, there are a few limitations of this study, including the small sample size and its focus on early adopters of one particular technology organization, which may not represent the broader early adopter population. Furthermore, the participants’ experiences were limited to prototype systems and internal team use only, leaving open questions related to real-world scalability and integration. As multi-agent generative systems become more prevalent, the human-agent interaction designs informed by the lessons learned from early

adopters will play a vital role in shaping how these systems are integrated into everyday workflows, ensuring that they are both useful and trusted by users.

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